FSD - Harnessing Gen AI: From Design to Code Optimization

Generative AI in Design Phase

Engage and Think

Imagine you are managing a design team assigned to create a mobile application for a travel company. The app aims to revolutionize how users interact with the company's products and services, incorporating personalized recommendations and intuitive user interfaces. As the design phase progresses, you recognize the complexity of the project and the need for a more creative approach and ideas.

How might Generative AI be leveraged to enhance the design process and streamline the development of complex features?

Learning Objectives

By the end of this lesson, you will be able to:

- Create architectural design for an application, consolidating design requirements and finalizing elements of the UI using Generative AI
- Utilize Generative AI tools to create the frontend of an application, maintaining consistency in coding
- Generate cards using Microsoft Designer to maintain visual and editorial consistency in the frontend
- Develop a working prototype of an application demonstrating the working of different UI screens and functionalities

Software Designing with Generative AI

What Is Software Development Lifecycle (SDLC)?

SDLC is a process used by software development teams to design, develop, and test software.
It defines the various stages or phases that are typically followed in the development of software.

Project planning

Requirement gathering and analysis

Designing

Implementation

Testing

Deployment

Support and maintenance

Design Phase in Software Development

The Design phase involves creating detailed architectural blueprints and user interfaces for the application or software and ensuring that functional and non-functional requirements are met.

Generative AI can be used in this phase to bring the requirements into a working prototype by making an attractive and user-friendly user interface of the application for all the functionalities.

Design Phase Activities

System and Architecture Design: Establishing the overall structure and technology stack of the application

User Interface and User Experience Design: Crafting the layout, wireframes, and interaction flow of the user-facing part of the application

Security and API Design: Defining security measures and creating APIs to ensure secure data flow and communication between system components

Prototyping and Documentation: Developing prototypes to validate design decisions and preparing comprehensive documentation

How Can Generative AI Help?

System and architectural design	It can suggest optimal architecture patterns, database structures and relationships based on system requirements, like recommending microservices or monolithic architectures.
User interface and user experience	It can generate UI wireframes and layouts based on user input or design guidelines, quickly creating interactive prototypes.
Security and API design	It can automatically generate API endpoints and documentation based on system requirements and data models.
Prototyping and documentation	It can also automatically create working prototypes and detailed design documents, including technical specifications, architecture diagrams, and data flow descriptions.

Popular Generative AI Tools for Design Phase

GitHub Copilot	AI-powered code completion tool providing real-time suggestions in the editor
Amazon Q	AI tool by Amazon delivering real-time code suggestions and completions
ChatGPT	Conversational AI by OpenAI capable of assisting with various tasks, including code generation

Popular Generative AI Tools for Code Generation

Copilot	Conversational AI-powered tool to provide real-time code suggestions, autocompletion, and error handling.
uizard	Uizard features multimodal capabilities, blending text, images, modular layouts, UI elements, and context for advanced AI applications.
snyk	Snyk helps proactively incorporate security by identifying potential vulnerabilities in software architecture, dependencies, and third-party components before implementation begins.

Industries Using Generative AI

Various industries globally are leveraging Generative AI technology for designing an intuitive and user-friendly experience in different operations, such as:

Healthcare	Automates coding for proprietary medical applications storing medical records and diagnosis, enhancing efficiency in healthcare operations
Finance	Streamlines tasks like fraud detection and generates personalized investment portfolios
Retail	Enhances customer experiences by creating personalized product recommendations
Software	Automates standard codes of various functionalities for the user interface and adapts according to the domain of the application

Business Benefits of Generative AI in Design Phase

Generative AI offers a range of capabilities to streamline and enhance the design process, including:

Faster time-to-market	Scalability	Error detection and debugging	Cost efficiency
Automates complex design tasks like creating system architecture diagrams, simulating design alternatives, and evaluating performance metrics, reducing manual iterations	Adapts to varying design requirements, optimizing architectures or microservices by simulating performance under different conditions, enabling easy scalability	Helps in identifying potential bugs and suggests fixes	Optimizes resource allocation and avoids unnecessary cost from design rework or inefficient system architecture

Generative AI in Design Phase: Example

Example

Task: Develop a basic signup system with user inputs

AI assistance: The GenAI tool generates a HTML code for user registration functionality

html Copy code

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Signup Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }
```

Generative AI: Architectural Design

What Is Architectural Design?

Architectural design in SDLC is the process of defining a structured solution that meets all the technical and operational requirements while maintaining quality and performance.

The key activities include:

01

High-level structure

Includes identifying the software components, modules, interfaces, and data flows

02

Technology stack

Involves selecting the appropriate technology stack, such as frameworks and libraries that aligns with the project's requirements

03

Architectural pattern and documentation

Identifies patterns like microservices, monolithic architecture, and layered architecture along with detailed documentation

Requirements of Architectural Design

It sets the foundational blueprint for a software system, guiding its development and evolution. The following are the key components:

01

Purpose and goals

Define the primary objective that aligns with business strategy

02

Functional requirements

Identify the essential function that the system must perform

03

Non-Functional requirements

Define quality attributes like performance, scalability, and more

Requirements of Architectural Design

04

Constraints

Recognize limitations in budget, time, technology, and resources

05

Stakeholder analysis

Gather input from end-users, clients, developers, and other stakeholders

Example: Generative AI for Architectural Design

Example

Prompt: Develop a basic architectural design for an ecommerce application

AI assistance: The GenAI tool generates a complete structure of the application with different layers in the application for both frontend and backend.

Designing an eCommerce application requires an architectural framework that supports scalability, security, flexibility, and user experience. Below is a high-level architectural design, breaking down key components of the system.

1. Presentation Layer (Frontend)

This layer includes the user interface and user experience elements through which customers interact with the application.

- **Technologies:** React, Angular, Vue.js, or plain HTML/CSS with JavaScript
- **Key Components:**
  - **Home Page:** Displays featured products, promotions, and navigation links.
  - **Product Catalog:** Lists products by category or search filters.
  - **Shopping Cart:** Displays items customers wish to purchase.

Example: Generative AI for Architectural Design

Example

Prompt: Develop a basic database structure for a finance application

AI assistance: The GenAI tool generates a complete DB structure of the application with different tables and their relationship.

1. Users Table

This table will store the information of all users who sign up for the finance application.

sql Copy code

```sql
CREATE TABLE Users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(100) NOT NULL UNIQUE,
    email VARCHAR(150) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

Demo: Generating Architecture Diagram Using Generative AI

Duration: 15 minutes

Demonstrate the process of generating architecture diagram using generative AI

You are tasked with designing the architecture for critical components of an e-commerce platform, focusing on enhancing functionality and user experience.

This demonstration reveals how generative AI, specifically a Software Architecture GPT, is employed to generate detailed architectures for a login page, add to cart page, payment page, and purchase page. Each component is carefully designed to ensure security, seamless integration, and efficient transaction handling, with the AI providing both textual descriptions and ASCII art diagrams.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
01_Generating_Architecture_Diagram_Using_Generative_AI

Generative AI for UI/UX

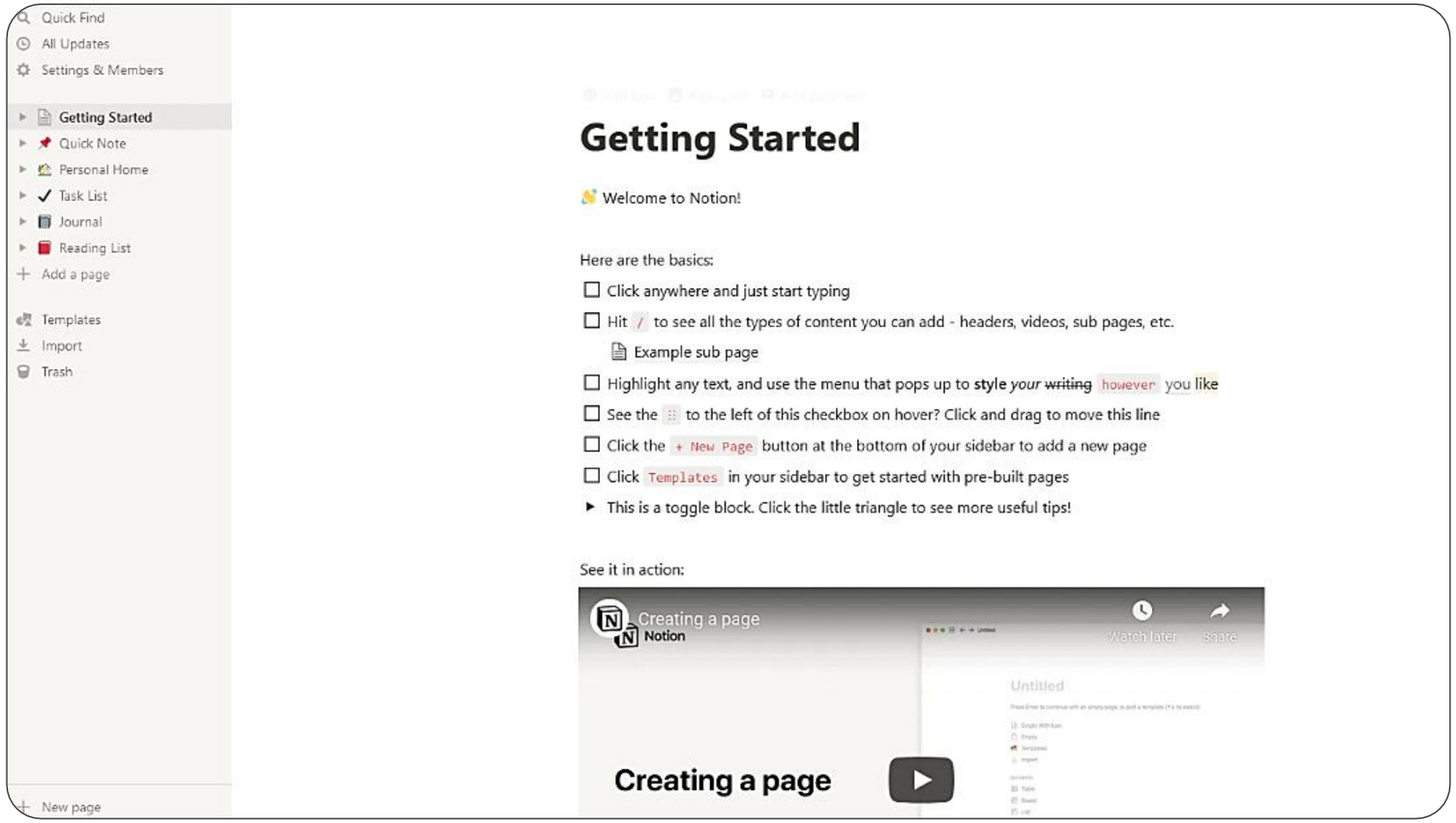
User Interface (UI) Design

It encompasses all the elements that enable users to interact with a product or service, including buttons, icons, spacing, typography, color schemes, and responsive design.

The goal of UI is to create a seamless and visually appealing interface that users find intuitive and enjoyable.

Example of a UI Design

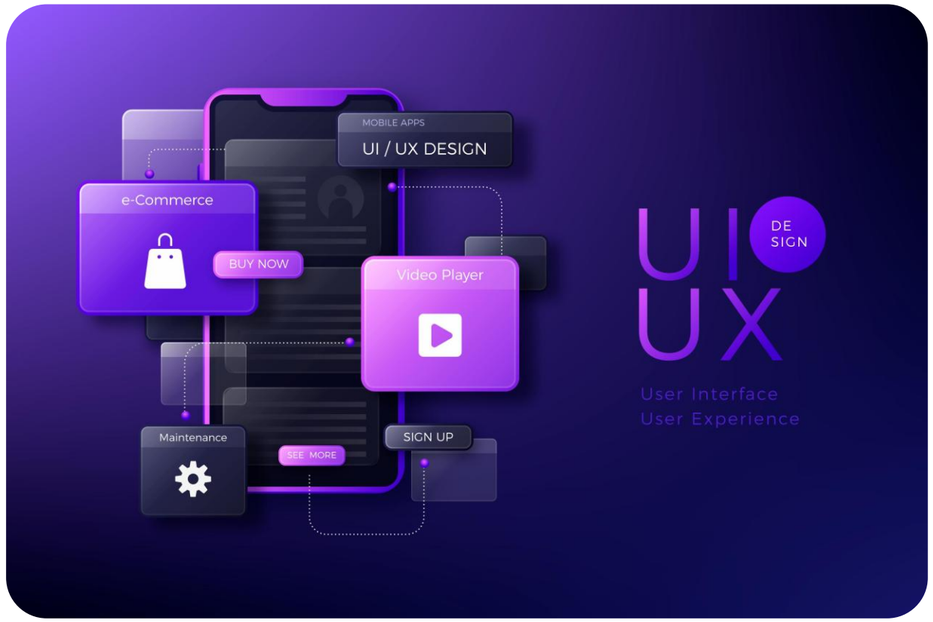
Notion's (productivity application) onboarding process begins on a clear and well-structured Getting Started page, providing instructions in easy-to-follow steps and a list to track and check off progress.



Source: https://www.notion.so/Getting-Started-10e9cc2258f380cfbef4f75b047f1c68

User Experience (UX) Design

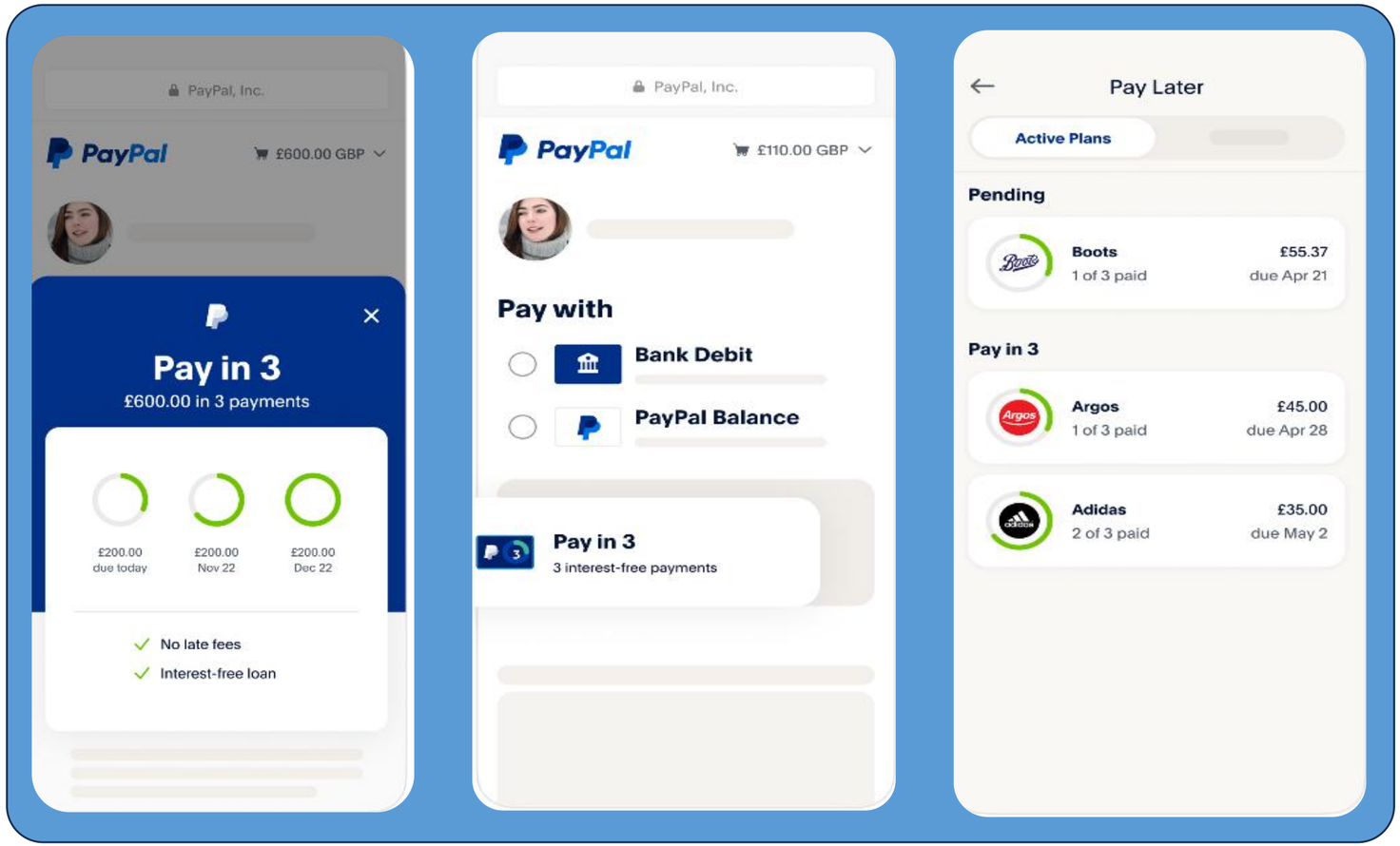
It is the process of enhancing user satisfaction by improving the usability, accessibility, and overall experience during interactions with a product.



It involves a comprehensive understanding of users and their needs to create a seamless journey through the product.

Example of a UX Design

PayPal (online payment platform) has simplified its website and mobile app experience, making all functions easy to find and use.



Interdependence of UI and UX

UI and UX are interdependent; one is about the look and the other about the feel.

UI cannot exist without UX	UI focuses on the tangible elements that users interact with, and the UX defines the context and efficiency of those interactions.
UI brings the user experience UX to life	UI acts as the bridge between the user and the digital environment, translating UX strategies into tangible interfaces.

Generative AI in UI/UX Design

It helps in the following stages of UI and UX design:

Idea exploration	Generates numerous design variations based on given parameters
Layout creation	Automates layout based on design principles and user preferences
Color schemes innovation	Analyzes color theory and trends to produce harmonious palettes
Typography enhancement	Analyzes typographic trends and user preferences to suggest suitable fonts and typography styles
Responsive design adaptation	Optimizes UIs for various devices by suggesting layout adjustments and responsive techniques

Generative AI in UI/UX Design

Image and icon generation	Assists in creating tailored images, icons, and graphics that enhance the design aesthetic
User experiences	Customizes interfaces to individual preferences for a more engaging and relevant experience
Content development	Ensures consistency across product descriptions, headlines, and interface copy
Usability insights	Forecasts user interaction and identifies usability challenges before live deployment
Design collaboration	Streamlines collaboration by automating feedback analysis and offers actionable insights

Demo: Generating UI Design Using Uizard

Duration: 15 minutes

Demonstrate the process of generating UI design using Uizard

You are responsible for developing the user interface (UI) for a grocery company's web application using Uizard. Your goal is to create a connected and stylish design.

This demonstration illustrates how Uizard AI facilitates the creation of a web application. Setting design preferences to 'Modern and Elegant' for the grocery application, which features various interconnected pages like a homepage, user profile, cart, and wishlist, Uizard efficiently produces a cohesive and visually appealing interface.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
02_Generating_UI_Design_Using_Uizard

Demo: Generating Cards Using Microsoft Designer

Duration: 15 minutes

DEMONSTRATION

Demonstrate the process of generating cards using Microsoft Designer

You are creating a greeting card for Freshbar, a company that specializes in milk products using Microsoft Designer to enhance its visual communication.

This demonstration details how Microsoft Designer is utilized to craft a unique greeting card that reflects the brand's identity. Starting from selecting a style from the card design options to customizing the content to reflect Freshbar's efficient supply chain and product range, the process is streamlined for ease and creativity.

Note: Refer to the demo document for detailed steps:
03_Generating_Cards_Using_Microsoft_Designer

Quick Check

You are a UI/UX designer working on a mobile app for an e-commerce platform. The team wants to improve the user interface to increase customer engagement and make product recommendations more intuitive. They decide to integrate Generative AI to assist in this process. How can Generative AI best assist you in improving the app's user interface to enhance the user experience?

A. By randomly generating new interface layouts without considering user data

B. By analyzing user interaction patterns and generating personalized UI design suggestions tailored to individual users

C. By fully automating the UI design process, eliminating the need for designer input

D. By automating A/B testing of UI designs without any user feedback

Personalization and Customization in UI/UX Design

Personalization in UI and UX Design

Definition

It involves dynamically adjusting content and interactions to align with users' preferences, behaviors, and past interactions.

Example

An online store gives you product recommendations based on your recent searches.

Customization in UI and UX Design

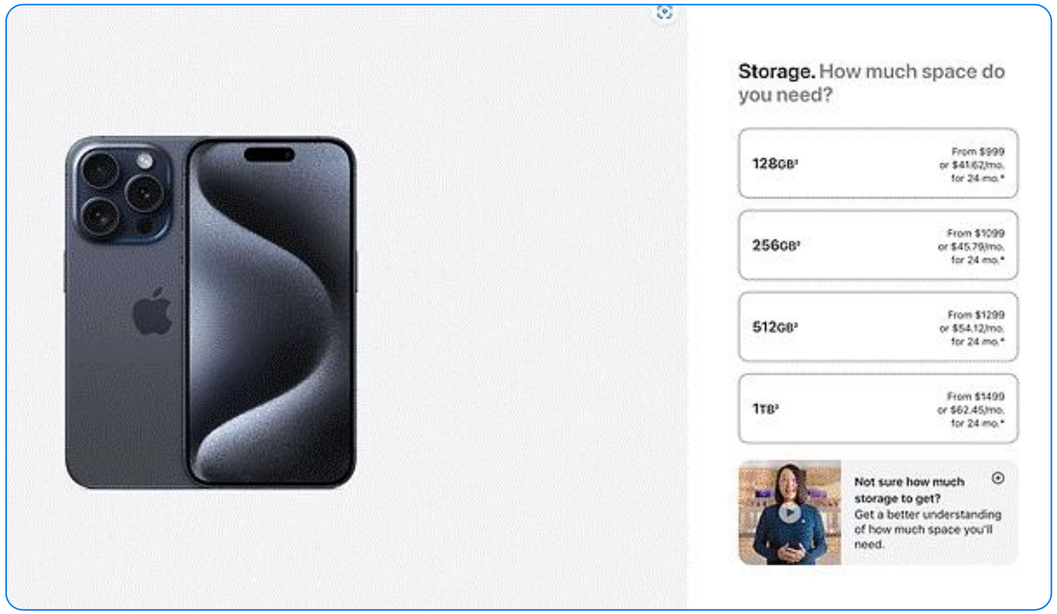
Definition

It refers to allowing users to modify certain aspects of the interface or functionality according to their preferences.

Example

A music app asks you to choose genres or artists you like to create your personal music feed.

Examples



Customization

Apple allows users to customize device types, colors, and specifications, providing them with a sense of ownership and personalization.

Personalization

Amazon provides personalized product recommendations based on users' browsing history, purchase history, and preferences.

Types of Personalization

Content personalization	It includes personalizing the content users see based on their preferences, past behaviors, or demographics to increase relevance and engagement.
Interface personalization	It includes adapting the visual elements of the UI, such as layout, theme, and navigation, to match the user's preferences and usage patterns.
Interaction personalization	It includes modifying the way the product interacts with the user, including personalized recommendations, adaptive user journeys, and tailored communication to enhance the user experience.

Example: Enhancing UX with Personalization and Customization

Scenario

E-commerce company ABC needed to create a compelling shopping experience to differentiate itself in the competitive market.

Solution

ABC implemented personalization and customization, offering tailored product recommendations based on user data and allowing users to adjust settings as per their preferences.

Outcome

Personalization and customization increased user engagement, satisfaction, and loyalty, leading to improved metrics like time spent on the platform and conversion rates.

Quick Check

Imagine you are the product manager for a music streaming application aiming to improve user satisfaction. In this context, which type of personalization would be crucial for adjusting the application's operational behavior and functionality to align with individual user habits and preferences?

A. Content personalization

B. Aesthetic personalization

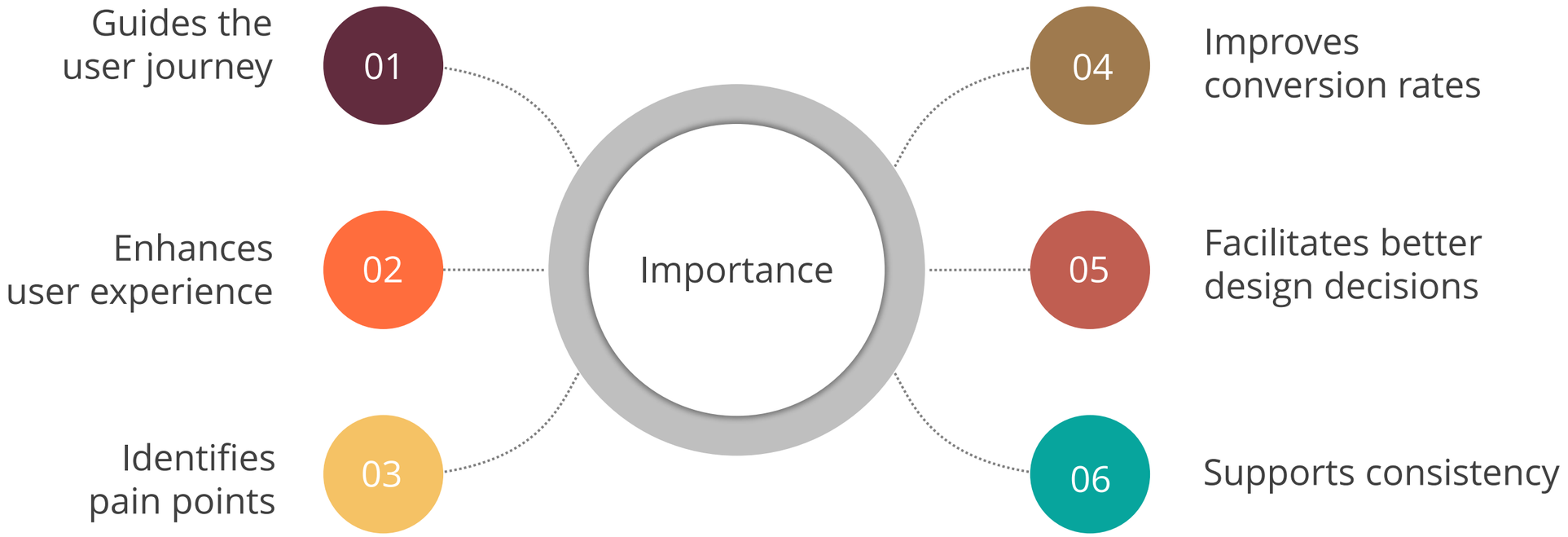
C. Interaction personalization

D. Visual personalization

Enhancing User Flow and Navigation

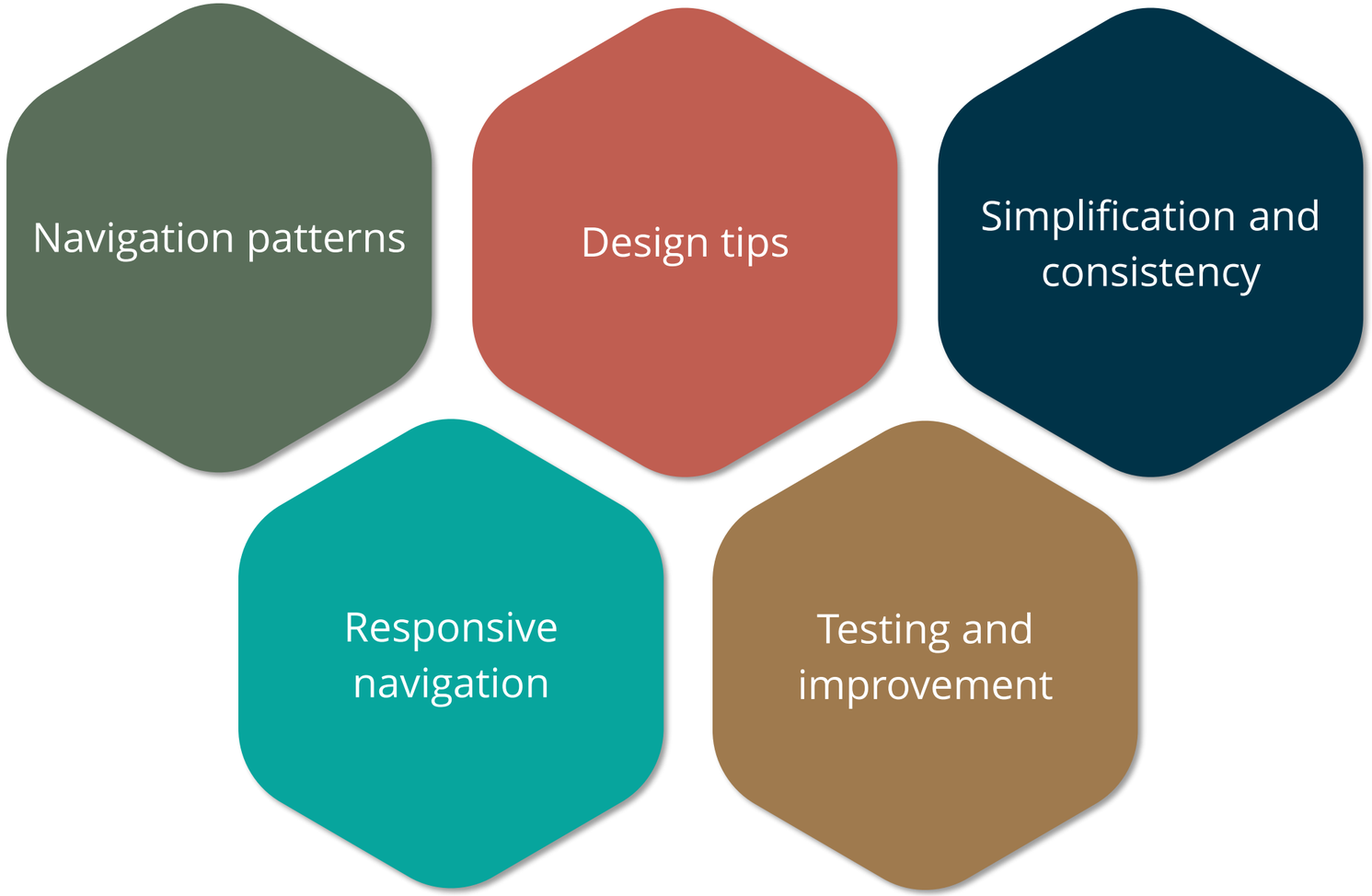
Importance of User Flow and Navigation in UI and UX Design

User flow defines the path users take to achieve their goals, and navigation systems guide users effortlessly through an application. The following points highlight their importance:



Techniques for Enhancing User Flow and Navigation through Design

Enhanced user flow and navigation in UI and UX design can significantly improve the user experience. Some of the techniques that help in user flow and navigation are:



Navigation Patterns

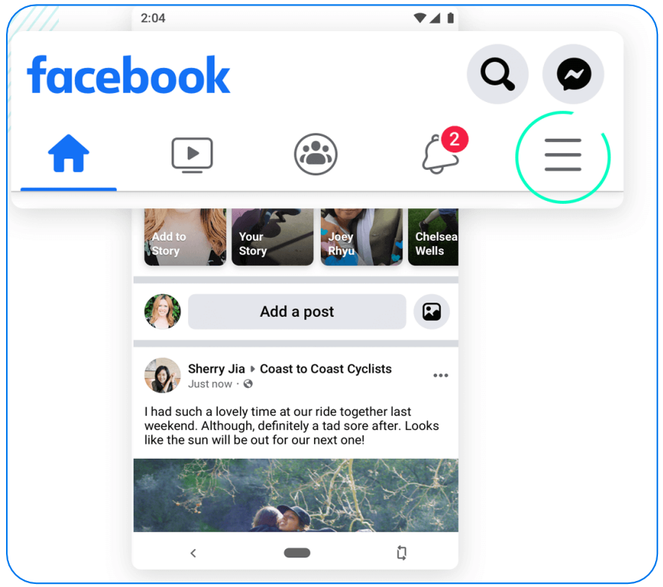
It establishes design strategies for organizing and presenting navigation elements within interfaces. It consists of the following patterns:

Tooltip pattern	Use tooltips for on-the-spot guidance on complex features
Spotlight pattern	Use a UI spotlight to direct attention during onboarding
Hamburger menu	Provide a compact menu on mobile devices to conserve space

Navigation Patterns

Vertical navigation menu	Employ vertical layouts for scalability and the inclusion of numerous item in customized menu
Progressive disclosure menus	Presents key elements first and then reveals more options as needed
Breadcrumbs	Offers a navigational trail for easy backtracking in intricate interfaces

Navigation Patterns



Example

The facebook.com image displays a hamburger menu with a top and bottom navigation bar.

Design Tips

It involves a collection of best practices and strategies aimed at improving the overall usability, functionality, and user experience of a product. These tips cover various aspects:

01

Enhance visibility

Use contrast, size, and placement to make navigation elements stand out

02

Simplify messaging

Use clear and descriptive navigation labels to ensure ease of use

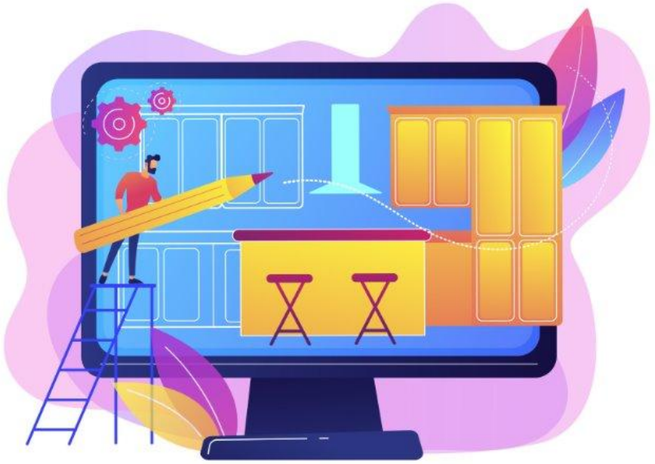
03

Location indicators

Use breadcrumbs and interactive walkthroughs to help the user know their location in the app

Simplification and Consistency

These are fundamental principles in the design phase to improve user experience and comprehension.



Simple layout
Reduce complexity to make navigation more intuitive

Consistent design
Maintain uniformity across the product to help users learn fast

Responsive Navigation and Testing

Design adaptable navigation that provides a uniform experience across all device types and screen sizes. The following image shows a responsive design that supports all devices.

Engage in continuous user testing to gather feedback on navigation effectiveness and implement required changes.

Demo: Generating an Attention Insight Using Figma

Duration: 15 minutes

Demonstrate the process of generating an attention insight using Figma

You are refining the user interface design for an e-commerce platform, focusing on grocery products, by utilizing Figma's Attention Insight plugin.

This demonstration explains how Figma facilitates the creation and evaluation of visual designs to effectively capture user attention. After loading images into newly created pages labeled Product Search and Profile, the process employs the Attention Insight plugin to generate heatmaps, focus maps, and contrast maps.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
04_Generating_an_Attention_Insight_Using_Figma

Wireframing

It is a visual guide that represents the skeletal framework of digital products. It simplifies the layout of content and functionality on a page according to user needs.

Importance of Wireframing and Prototyping in Design Workflows

01

Enhances design and precision

Provides a clear layout for stakeholders and enables issue identification

02

Enhances efficiency

Supports rapid exploration of design options and quick iteration

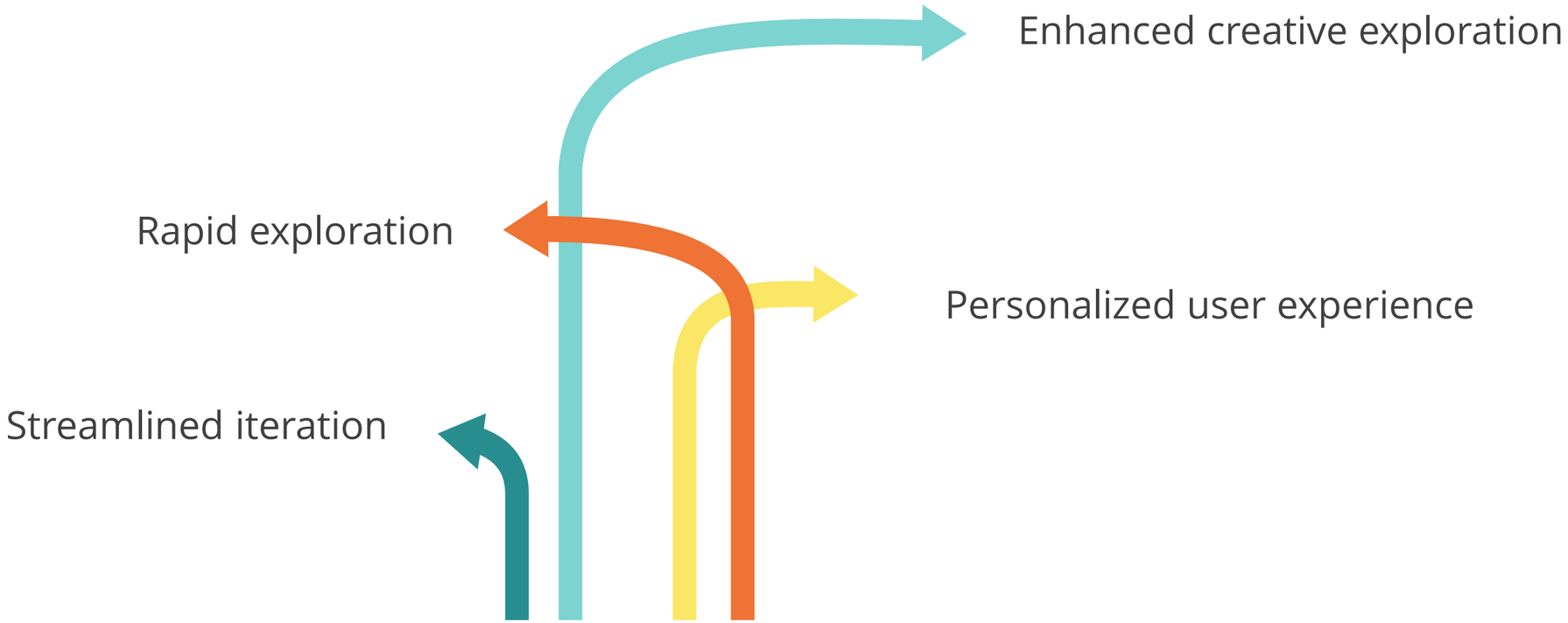
03

Facilitates user testing

Assesses usability and user flow for direct feedback on design's effectiveness

Generative AI Accelerates Wireframing and Prototyping

It accelerates the wireframing and prototyping stages of UI and UX design by automating and optimizing various processes. AI helps in the following tasks in the process:



Demo: Implementing Generative AI in Framer for E-commerce Prototyping

Duration: 15 minutes

You are developing a responsive web design for Fresh Start Groceries, an e-commerce platform, using Framer to generate and enhance user interfaces.

This demonstration explains how Framer is leveraged to create a dynamic and responsive webpage for an online grocery store. Starting with generating a homepage that features a navigation bar, search field, and product sections like 'Featured Products' and 'Deals of the Day', the process involves using AI prompts to design subsequent pages for categories and product details.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
05_Implementing_Generative_AI_in_Framer_for_E-commerce_Prototyping

Demo: Generating Code for an E-commerce Application Web Page

Duration: 15 minutes

You are tasked with developing a set of web pages for an e-commerce application, including login, sign-up, and product display pages.

This demonstration will guide you through developing a website's login, sign-up, and product display pages. You'll create user authentication interfaces that allow seamless navigation between the login and sign-up pages and design an attractive product display page using HTML and external CSS.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
Demo_06_Generating_Code_for_an_E-commerce_Application_Web_Page

Demo: Generating E-commerce Product Module Using JavaScript

Duration: 15 minutes

You are tasked with creating an e-commerce product module using JavaScript, enabling users to add, store, and view product details with seamless navigation and a well-structured interface.

This demonstration details how to set up the project structure, implement product management pages, link these pages, and test the entire module. The steps involve creating HTML pages for adding and viewing products, along with JavaScript for managing product data and interactions.

DEMONSTRATION

Note: Refer to the demo document for detailed steps:
Demo_07_Generating E-commerce Product Module Using JavaScript

Quick Check

Imagine you are a mobile app designer tasked with optimizing screen space on small devices. In mobile app navigation design, which navigation pattern takes precedence over others to conserve screen space by concealing the primary navigation menu?

A. Tooltip pattern

B. Spotlight pattern

C. Breadcrumbs

D. Hamburger menu

Key Takeaways

- Architectural design serves as a conceptual model that outlines the structure of a system and delineates how its components interact within a given framework.
- UI and UX are closely interrelated, with UI focusing on the visual aspects of design and UX emphasizing the overall feel and user interaction.
- Generative AI brings transformative capabilities to every stage of UI/UX design, significantly boosting creativity and efficiency in the design process.
- Personalization involves dynamically adjusting content, features, and interactions to suit the unique characteristics, behaviors, and histories of individual users.
- Wireframing acts as the preliminary sketch that establishes the structure and layout of a website or application.

Guided Practice

Overview **Duration: 30 minutes**

In this project, we delve into the practical application of generative AI prompts to expedite the frontend development process for a grocery e-commerce website.

By leveraging ChatGPT, we aim to streamline the creation of essential web pages such as contact, shop, and cart pages. The goal is to ensure seamless site navigation, facilitated by XAMPP as the local server environment and PHP for backend scripting.

Task:

a) Create a contact page for the grocery website using ChatGPT
b) Generate a cart page for the grocery website through ChatGPT
c) Generate CSS script to fix header and footer

Practice Project

Creating a Frontend Starter Code for a Taxi Ride Booking Application Using Generative AI

The objective is to develop frontend starter code for the Book_A_Taxi ride booking application using Generative AI tools. The primary task is to use AI-driven prompts to establish the foundational code structure for the website.

By participating in this project, learners will gain hands-on experience in frontend development, specifically in building a taxi ride booking application.

Task:

1. Generate a frontend starter code for a taxi ride booking application
2. Create additional pages for the website, including About Us, Services, and Contact pages
3. Link all pages together to ensure seamless navigation within the website
4. Add a new page for booking a ride for various services and link it with existing pages
5. Implement basic form validation for the booking form on the website

PRACTICE PROJECT

Q&A